Divyang Deepak Palshetkar

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Summary

Environmental engineer and fire scientist (**Dept. Rank 1**, IIT Jodhpur) specialising in **coupled fire–atmosphere–chemistry modelling** with WRF-Chem-Fire / WRF-SFIRE, wildfire spread dynamics, aerosol radiative feedback, and fire–weather interactions in complex Himalayan terrain. Proficient in **Python**, **R**, **ArcGIS/QGIS**, **NetCDF/GRIB** spatial pipelines, and statistical analysis. Independently designed and executed the **first high-resolution coupled WRF-Chem-Fire simulation** of a major Indian Himalayan wildfire episode, producing peer-quality quantitative results across fire behaviour, air quality, and atmospheric dynamics.

Education

B.Tech–M.Tech Dual Degree, Civil & Infrastructure (Environmental) Engineering *2021 – 2026 (Expected)*

- **Indian Institute of Technology (IIT) Jodhpur** | B.Tech CGPA: **8.3 / 10** | M.Tech CGPA: **9.5 / 10**
- **Department Rank: 1**
- **Key Coursework:** Air Pollution & Control (A), Climate Change & Impact (A), Geo-informatics / GIS (A), AI/ML in Infrastructure Engineering (A), Environmental Engineering (A), Water Resources Engineering, Fluid Mechanics (A-)

Core Skills

Fire & Atmospheric Modelling: WRF-Chem-Fire / WRF-SFIRE, Rothermel fire spread model, two-way fire–atmosphere coupling, level-set perimeter tracking, Grid-FDDA data assimilation, MOZCART photochemistry, GOCART aerosol module, RRTMG radiation, PBL dynamics, fire plume-rise (Paugam scheme), biomass burning emission modelling

Spatial Data & GIS: ArcGIS Pro, QGIS, NetCDF / GRIB / raster processing (Python Xarray, NCO), NASA FIRMS / MODIS / VIIRS active fire & burn scar data, ERA5 reanalysis, DEM / terrain analysis, MODIS land cover

Programming & Analysis: Python (NumPy, Pandas, Xarray, Matplotlib, SciPy, Scikit-learn), R (spatial analysis, statistical modelling), Bash scripting, process automation, ML for environmental systems, Git / GitHub

Fire Science & Risk: Fire behaviour analysis (wind–slope–fuel), fire emission inventories (FINN v2.5, EDGAR-HTAP), aerosol optical depth & radiative forcing, air quality assessment ($PM_{2.5}$, PM_{10}, BC, OC, O_3, NO_x), fire weather indices, fire risk quantification

HPC & Computing: Linux HPC cluster, parallel WRF execution, large-scale simulation workflow management, technical documentation & scientific reporting

Research & Projects

M.Tech Thesis: Coupled WRF-Chem-Fire Simulation of the 2024 Uttarakhand Wildfire *2025 – 2026*

IIT Jodhpur | Supervisor: Dr. Amit Sharma | WRF-Chem-Fire, Python, NetCDF, ArcGIS, ERA5, NASA FIRMS

- Delivered the **coupled WRF-Chem-Fire simulation** of the April 2024 Champawat–Purnagiri wildfire (Uttarakhand), a $\sim$414 km^2 burn event at **1 km nested resolution** over the active fire zone.
- Implemented **Rothermel fire spread** via level-set algorithm with **two-way fire–atmosphere coupling**, **MOZCART** photochemistry, GOCART aerosol module, and RRTMG radiation; configured Anderson NFFL Category 9 (hardwood litter) fuel properties calibrated to Indian Himalayan forest conditions.
- Applied **Grid-FDDA nudging** (ERA5): wind speed NMB reduced by **29–38%**, IOA improved by **40–44%** over free-running WRF validating the meteorological baseline critical for accurate fire spread.
- **Fire emission impact:** Peak domain-averaged $PM_{2.5}$ reached **71 µg m^{-3}** (exceeding NAAQS); BC ↑**9**×, OC ↑**10**×, NO_2 ↑**2.5**×; $PM_{2.5}$ model bias improved from **NMB −19.2% → +6.4%** upon activating fire emissions.
- **Aerosol radiative feedback:** SWDOWN reduced **15–25 W m^{-2}**; PBLH suppressed **20–50 m**; near-surface $PM_{2.5}$ enhanced **5–12 µg m^{-3}** (self-amplifying accumulation feedback, $\sim$7% gain); O_3 depleted **1–3 ppbv** via UV photolysis attenuation; fire spread insensitive (<1.3% burnt area change), confirming wind/terrain dominance.
- **Fire spread dynamics:** Characterised **diurnal pulsing** behaviour, two distinct peak spread periods (18 & 19 April); peak ROS **≈6.0 m s^{-1}**, areal growth **>1.5 km^2 hr^{-1}**; total burnt area **≈470 km^2** in ≈92 hours. Performed topographic wind channeling analysis (Froude number) explaining multi-lobed fire perimeter geometry.
- Processed **multi-source spatial datasets** in NetCDF/GRIB: ERA5 (IC/BC), NASA FIRMS MODIS/VIIRS (ignition points), EDGAR-HTAP emissions, FINN v2.5 fire inventories, CAM-Chem chemical BCs, MODIS MCD12Q1 land cover.
- Validated against **CPCB CAQMS** real-time observations ($PM_{2.5}$, PM_{10}, O_3, NO_2, wind, temperature) using NMB, nRMSE, Pearson r, IOA metrics; domain resolution sensitivity analysis (1 km vs. 3 km).

Wildfire Risk Mapping & Fire Weather Spatial Analysis — Python, GIS *2024 – 2025*

IIT Jodhpur | Python, ArcGIS, QGIS, ERA5 NetCDF, Remote Sensing

- Developed automated Python pipelines for **ERA5 NetCDF** ingestion and processing (temperature, wind, RH, precipitation) to derive **fire weather indices** and identify extreme pre-monsoon fire weather periods over Uttarakhand.
- Integrated **MODIS land cover, DEM, and meteorological rasters** in ArcGIS/QGIS to produce **spatially explicit fire hazard exposure maps** across fuel type and terrain gradients.
- Applied **statistical trend analysis and time-series visualisation** on historical fire season data to quantify drivers of wildfire intensification in the Indian Himalayan Region.
- Generated **automated model validation scripts** comparing WRF-Chem-Fire outputs against NASA FIRMS satellite detections and CPCB ground observations.

Urban Flood Risk & Drainage Modelling — HEC-RAS, GIS *2024*

IIT Jodhpur | HEC-RAS, Civil 3D, ArcGIS

- Designed urban stormwater conveyance systems using HEC-RAS and GIS, improving drainage capacity by **27%** and reducing earthwork by **19%**; produced construction-ready drainage drawings.

Work Experience

Civil Engineer, EEKI FOODS March – May 2025

- Developed GIS-based **flood risk mitigation** strategies through hydraulic and spatial analysis, reducing hazard zones by **32%**; produced terrain models and drainage designs using Civil 3D.

Engineering Intern, Jal Jeevan Mission (Ministry of Jal Shakti) May – July 2024

- Conducted GPS/QGIS mapping across 40+ sites; developed 3D hydraulic models for recharge structure evaluation, improving groundwater replenishment efficiency by **24.6%**.

Research Intern, AICOE Project, Ministry of Education May – July 2024

- Built annotated traffic datasets supporting **AI-powered** infrastructure digital twin modelling; developed automated data validation pipelines with interdisciplinary teams.

Achievements & Positions of Responsibility

- **Department Rank 1** — B.Tech–M.Tech Dual Degree, Civil & Infrastructure Engineering, IIT Jodhpur
- **M.Tech CGPA 9.5 / 10** — M.Tech thesis represents the high-resolution coupled WRF-Chem-Fire study of a major wildfire episode in the Champawat–Purnagiri region, Uttarakhand
- **Festival Chief, Ganeshotsav** — Led 200+ volunteers across cultural operations and campus logistics
- **Head, Public Relations, EDIFICIO** — IIT Jodhpur Civil Engineering Festival